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V-shaped PSF for 3D imaging over an extended depth of field in wide-field microscopy

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Single-shot 3D optical microscopy that can capture high-resolution information over a large volume has broad applications in biology. Existing 3D imaging methods through point-spread-function (PSF) engineering often have limited depth of field (DOF) or require custom and often complex design of phase masks. We propose a new PSF approach that is easy to implement and offers a large DOF. The PSF appears to be axially V-shaped, engineered by replacing the conventional tube lens with a pair of axicon lenses behind the objective lens of a wide-field microscope. The 3D information could be reconstructed from the single-shot image through a deep neural network. We simulate the 3D image formation and reconstruction process in a wide-field microscope of 10× magnification. The V-shaped PSF shows an excellent 3D resolving capability, with a < 2.5 μm lateral resolution and ~15 μm axial resolution over a DOF of ~350 μm at a 550 nm wavelength. Compared to other popular PSFs designed for 3D imaging, the V-shaped PSF is simple to deploy while offers high 3D reconstruction quality over an extended DOF.

In biological imaging, capturing detailed information over large 3D volumes is essential for understanding complex structures and processes. High-speed volumetric imaging is crucial in many applications, such as studying brain activity in large-scale neural circuits[1, 2] or visualizing macroscopic cleared samples[3]. Conventional microscopes, limited to imaging a single axial plane at a time, are unsuitable for these tasks. Spatial multiplexing techniques can image different depths at once by converting their images to different cameras[4] or different lateral positions on the camera[5, 6]. Still, these approaches require complex optical setups and have an inefficient photon use. Extended Depth-of-Field (DOF) techniques with elongated point spread function (PSF) in the axial direction can capture volumetric information at once[7-10]. However, common PSFs, such as low numerical aperture (NA) Gaussian beam and Bessel beam PSFs[7-9], lack axial variance and cannot distinguish features at different depths.

PSF engineering[11] offers a powerful method for single-shot 3D imaging by shaping the PSF to stay spatially confined while varying its profile along the depth, typically by an amplitude[12] or phase mask[11, 13-15] at the Fourier plane of the microscope. Techniques like astigmatic PSF[16], double helix[13, 14], and single helix[15] can capture 3D information in one acquisition. The tetrapod PSF[17] is specifically optimized for 3D imaging over a user-defined DOF. Light-field microscopy[18], with a microlens array at the image plane, also introduces axial variance to the PSF for 3D imaging. However, these methods are often designed for applications with a limited axial range, or require custom, and often complex design of phase masks. Imaging in high resolution over a large depth of field remains a general challenge.

In this letter, we report a V-shaped PSF that could resolve 3D information in high resolution over an extended DOF in wide-field microscopy. The V-shaped PSF is formed by two spatially separated and angled Bessel beams. A single object is thus imaged as two spatially displaced copies on the 2D image plane, with a separation distance proportional to the object depth in the 3D volume. Crucially, each lobe of the PSF stays spatially confined over an extended DOF, making it possible to resolve 3D features over a large DOF. Compared to many other PSFs engineered for 3D imaging, the V-shaped PSF is easy to implement and could achieve a higher 3D reconstruction quality over an extended DOF range. We note that a self-bending PSF, which also appears in V-shape, has been demonstrated previously through a spatial light modulator [19]. It requires a complex optical setup, precise polarization tuning, and system calibration. Instead, our PSF could be generated in wide-field microscopy by replacing the tube lens with a pair of laterally displaced axicon lenses in the collection path (Fig. 1), or by adding another tube lens and the axicon lenses (Supplementary Note 1, Supplementary Fig. S1). The volumetric object is projected into a single 2D image. Algorithms such as Richardson-Lucy deconvolution or deep neural networks can then reconstruct the 3D object. As a proof of concept, we designed the double-axicon lens for a 10× (0.25 NA, with ~20 μm native DOF) microscope objective lens to form the V-shaped PSF at 550 nm wavelength. Our simulations show that the microscope can achieve good lateral and axial resolutions, <2.5 μm and ~15 μm, respectively, over a ~350 μm DOF. Using a deep neural network, we reconstruct 3D objects in high quality with high intersection over union (IOU) against the ground truths. We show that the V-shaped PSF achieves high 3D reconstruction quality over an extensive DOF, comparable to the numerically optimized PSF for 3D imaging, while offering a much simpler design principle and implementation.

Double-axicon lens to generate V-shaped PSF

The V-shaped PSF is formed by two tilted Bessel beams, each with a central lobe that remains spatially confined over a long propagation

distance. Such a PSF is generated by replacing the tube lens in the collection path of a wide-field microscope with a pair of axicon lenses positioned adjacently (Fig. 1a, Supplementary Note 2). Alternatively, one could also add another similar tube lens and a pair of axicon lenses, between the original tube lens and camera (Supplementary Note 1, Supplementary Fig. S1). In the case shown in Fig. 1a, the pupil distance (PD) between the axicon lenses and the pupil plane of the objective lens is selected to achieve a high transmission efficiency over the extended DOF while maintaining a low distortion (±3% at the edge of a 700 μm×700 μm FOV over a 350 μm DOF). The nominal focal length of the axicon lens can be expressed as f≈R/[2(n-1)α], where R is the radius of the axicon, n is the refractive index of the lens material, and α is the tilting angle of the conical shape[9]. f can be set to be the same as that of the tube lens to be replaced. We set the diameter of the axicon lens to be the same as that of the pupil of the microscope and the separation of the two axicon lenses to be the radius of the axicon lens to achieve a high transmission efficiency while minimizing the material cost (Fig. 1b). An iris could be set at the axicon plane so only the light passing through the two axicon lenses could reach the detector (with ~92% transmission). A camera is placed at the nominal rear focal plane of the double axicon lens to collect the two copies of images with a separation dependent on the axial depth of the object.

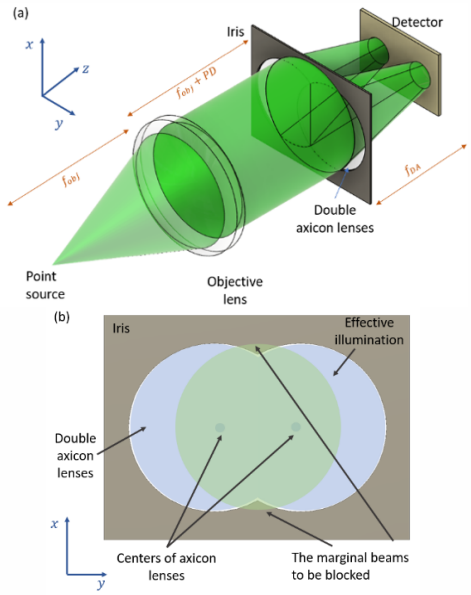


Fig. 1. Double-axicon lenses to generate the V-shaped PSF. (a) Schematic of the 3D extended depth of field microscope composed of an objective lens and double-axicon lenses (drawing not in scale). An iris is placed in front of the double-axicon lenses to reject the marginal beams. The double-axicon lenses serve as a tube lens. The detector plane is placed at the nominal rear focal plane of the double axicon lens. A V-shaped PSF is formed where the separation of the two lobes of the PSF changes with respect to object distance. (b) Front view of the double-axicon lenses and illumination beam.

We paired the double-axicon lenses with a 10× magnification microscope objective lens (0.25 NA). We set the effective focal length of the objective lens to be $f_{obj} = 18\,mm$, and the pupil diameter to be $D = 2 \cdot NA \cdot f_{obj} = 9\,mm$. The two axicon lenses' designed focal length and diameter were then set to $f_{DA} = 180$ mm and 9 mm, respectively, to create the 10× magnification and maximize light transmission. The two axicon lenses' centers were separated by 4.5 mm. Their small overlap creates some asymmetricity of the shape of each lobe of the PSF and slightly changes the focal length of the double axicon. Still, it has little impact on the 3D resolving capability and DOF.

System characteristics

We investigated the performance of this imaging system through a ray-tracing algorithm (Zemax OpticsStudio). We placed an object point source along the optical axis and monitored the PSF on the detection plane (Fig. 2a-b). As the axial position z of the point source varies, the separation $2l$ of the PSF's two lobes changes approximately linearly (Fig. 2c), in a ratio of $\Delta(2l) / \Delta z \approx 2.62$, where l is the lateral distance between each PSF lobe and the optical axis. In this manner, the PSF observed at the detector plane adopts the V-shaped form as the depth of the objects varies (Fig. 2a inset). The NA of each PSF lobe is approximately 0.125 and 0.25 in the y- and x-direction, respectively (Fig. 1), based on the geometry of each axicon. Over an axial distance of 350 μm of the point source, the Full Width at Half Maximum (FWHM) of each lobe of the PSF is <25 μm (Fig. 2d) in the y direction at the detector. As the system magnification is 10×, this corresponds to <2.5 μm of lateral resolution on the object plane. We defined the axial resolution as the minimum displacement of the point source in the axial direction needed to distinguish a noticeable difference in each lobe at the detector plane along the y-direction. According to the Rayleigh criteria, the axial resolving power was calculated to be ~15 μm.

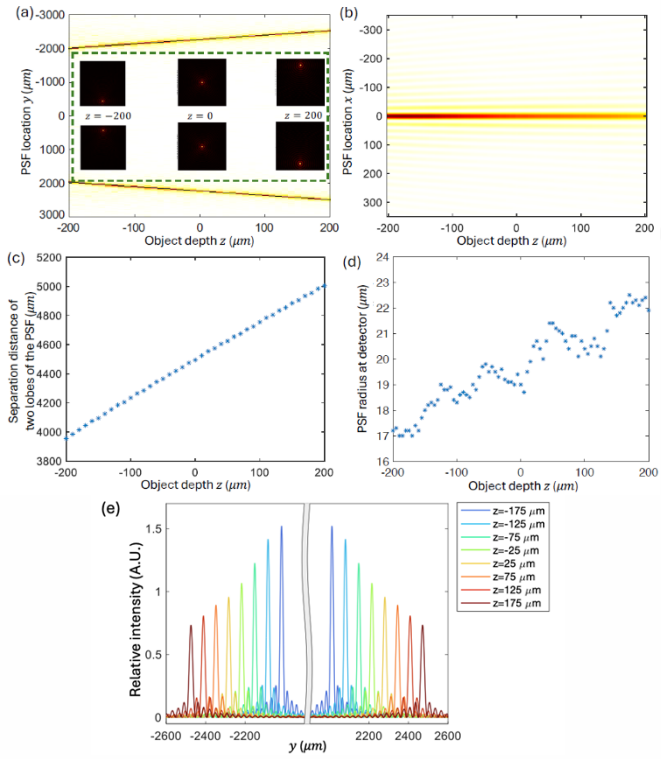


Fig.2. (a-b) Simulation of the V-shaped PSF. Illustration of the two PSF lobes imaged by the camera versus the depth of the on-axis object point source in the (a) yz plane and (b) xz plane intersecting the maximum value of the PSF. Inset of (a), intensity profile of the two main lobes of the PSFs in the xy plane for three different object depths. (c) Separation of the two PSF lobes versus the depth of the on-axis object point source. (d) Radius of the PSF main lobes against the object depth. The discontinuity quantization effect is due to the pixelization of the detector and precision in the numerical simulation. (e) The lateral PSFs (along y direction) at different object depths within the 350 μm DOF. The intensity of each PSF is normalized with respect to the PSF at the nominal focal plane. As the object point source moves towards the objective lens, more marginal beams are blocked by the iris at the axicon plane, resulting in a reduced transmission power.

As the point source moves towards the objective lens, the iris blocks a larger portion of the marginal beams from the source, leading to the vignetting and a decrease of intensity in the PSF (Fig. 2e). Within ±175 μm axial offset from the nominal focal plane, the minimum value among all the peak intensities of the PSF at different depths is ~50% of the maximum value of all the peak intensities. Using this standard, the DOF could be defined as 350 μm. Such a DOF is much larger than the native DOF of the objective lens (~20 μm), or when two convex lenses with the same focal length are used instead of the two axicon lenses (<30 μm DOF to keep a similar lateral resolution on x and y direction, simulated by OpticsStudio). We note that while Bessel beams provide an extended DOF, the central lobe contains significantly less energy as the number of rings increases, all with a uniform distribution [20, 21]. The lower energy density in the outer rings limit their effectiveness for fluorescence detection. Therefore, we expect that primarily the central lobe and those rings exceeding the camera's detection threshold will contribute meaningfully to object detection and reconstruction. We further studied the properties of the V-shaped PSF when the double axicon lens is shifted laterally or tilted, which mimics the real experiment condition (Supplementary Note 3, Supplementary Fig. S2-S5). Based on the 3D PSF, we proposed approaches to finely align the double axicon lens in the microscope.

3D Image Reconstruction

The microscope with the V-shaped PSF projects the 3D volumetric object space into a 2D image on the camera. With prior knowledge of the PSF (Fig. 2a), we can reconstruct the 3D object volume from a single 2D image. Conventional reconstruction algorithms (e.g., Richardson-Lucy deconvolution, fast iterative shrinkage-thresholding algorithm [22], etc.) are time-consuming and computationally intensive for highly compressed images or when the number of reconstruction pixel/voxel is large. Deep learning architectures, once trained, could offer faster run times and can learn regularization terms to enhance reconstruction quality. Here, we adopted a deep neural network based on Hadamard-Nets [23-26], which has proven robust and efficient for object reconstruction. The network comprises a learnable deconvolution module which reconstructs all the axial planes at once, and a 3D U-Net which refines and fuses the reconstruction from different depths (Supplementary Note 4, Supplementary Fig. S6). The core of the deconvolution module is a physics-informed multi-Hadamard-Net [26], where each Hadamard-Net learns the deconvolution kernel for a specific axial depth in frequency domain. The network takes the Fourier transform of the measured image as input and broadcasts it to N pairs, where N is the number of axial planes to be reconstructed. Each pair contains the real and imaginary components, which are respectively elementwise multiplied with a pair of learnable Hadamard kernels. Each pair of learned Hadamard kernels extracts object features from one axial depth. The product is then inverse-transformed to the spatial domain and stacked into a 3D volume with N depths. In the fusion module, a 3D U-Net is used to post-process the volume to suppress the background and remove duplicated out-of-focus features that appear on multiple depths. The fusion network outputs an N-depth 3D volume. As a proof of concept, we set N=8 here.

We took neuronal imaging in transparent samples as an exemplary application (i.e., imaging the neuronal activity in zebrafish *in vivo* or the neuronal network in 3D cleared samples). We constructed a database for object planes with randomly distributed neuronal cell bodies in 3D (each being ~8 μm in diameter; see Fig. 3a as an example). To resemble the 2D image as would be taken in the experiment, we convolved the object at a certain depth with the PSF (estimated as spatial invariant) generated by Zemax OpticsStudio at the corresponding depth with calibrated magnification, added with 5% Gaussian noise (related to amplifier noise and read noise of the camera) and 5% Poisson noise (related to shot noise). We then summed images from different depths into a single image (Fig. 3b), where the same cells appear to be imaged twice due to the V-shaped PSF, with their lateral separation depending on the depth of the cells. The single images and the corresponding object planes at individual depths were used as training pairs to train the neural network for reconstruction at the different depths.

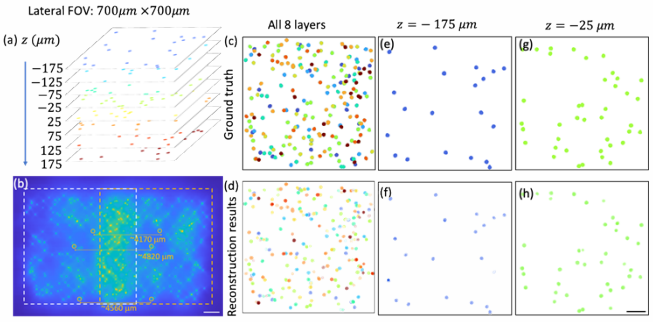


Fig. 3. 3D imaging and reconstruction of 8 evenly distributed object planes with neuronal cell bodies in each plane, across 350 μm DOF. (a) Ground truth of the 8 evenly distributed object planes within 350 μm DOF in the axial direction, with different colors indicating neuronal cell bodies at different depths. The FOV of each plane is 700 μm × 700 μm and the neuronal cell bodies have an average diameter of 8 μm. (b) Image of the 8 object planes through the V-shaped PSF at the camera. The yellow circles represent examples of double images from the same neurons. The separation of the two images is a function of the object depth. The two boxes in dashed line indicate the sensor regions for the images from the two PSF lobes, which could overlap if the FOV of the individual PSF lobe is large. (c) Same as (a), but with all 8 planes projected into a 2D view. (d) Reconstruction results at the 8 different planes, visualized in the 2D projection view. (e) Ground truth and (f) reconstruction at z=-175 μm plane. (g-h) Same as (e-f), but for z=-25 μm plane. In (d), (f) and (h), the gray-scaled color shows the relative intensity among the neurons in the reconstruction. Scale bar: (b) 1 mm; (h) 100 μm.

We evaluated the reconstruction performance for a volume with 8 evenly distributed object planes within the 350 μm DOF (Fig. 3) to simulate the multiplane light-sheet illumination where a set of selected planes are illuminated [27]. Each object plane has a size of 700 μm×700 μm. The neuronal cell bodies were well reconstructed in each plane with an averaged IOU over all test datasets between the ground truth and the reconstructed image of 0.65. The reconstruction time for 8 planes is ~35 ms (NVIDIA GeForce RTX 2080 Ti). While only selected planes with separation distances larger than the axial resolution were illuminated (by light sheet illumination) in this example, our reconstruction algorithm could handle the case where the entire volume is illuminated (Supplementary Note 5, Supplementary Fig. S7). Our deep neural network, while trained for a small range of neuronal density, could work for a wider range of neuronal density though in a reduced performance (Supplementary Note 6, Supplementary Fig. S8-S9).

We also benchmarked the reconstruction performance of the V-shaped PSF over other popular PSFs engineered for 3D imaging with the same neural network architecture (Fig. 4, Supplementary Note 7, Supplementary Fig. S10-S12). These PSFs include the conventional diffraction-limited PSF (DL-PSF), the astigmatic PSF[16], the double-helix PSF[13, 14], and the tetrapod PSF[17]. All of the PSFs are designed under the same NA and aperture size. The tetrapod PSF is obtained by numerical optimization of the Cramer Rao Lower Bound (CRLB) [11, 17] over the DOF of ± 175 μm, matching that of the V-shaped PSF.

Using the same methodology in Fig 3, we assessed the reconstruction quality for a volume with 8 evenly distributed object planes within the 350 μm DOF. Each plane has an FOV of 700 μm×700 μm, and contains 20~40 neurons. Though the reconstruction quality of the V-shaped PSF

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is not the best around the nominal focus, it outperforms the DL-PSF, the astigmatic PSF and the double-helix PSF over the extensive DOF (Supplementary Fig. S12), attributed to its tight focusing. Interestingly, the V-shaped PSF performs similarly to the tetrapod PSF, which is specially optimized for this DOF. Crucially, V-shaped PSF does not require complex numerical optimization; it can be designed through an intuitive top-down approach and implemented easily.

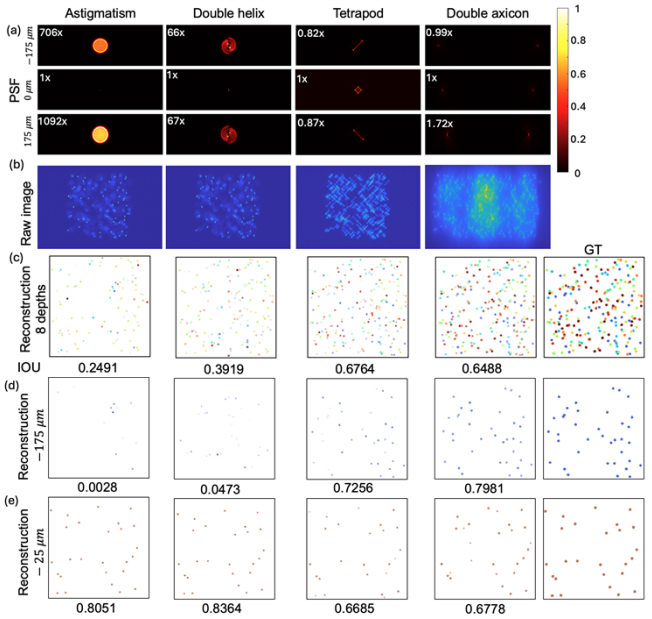


Fig. 4. A comparison of different PSFs (astigmatic, double-helix, tetrapod, and V-shaped) in image reconstruction performance through a neural network with the same architecture. The samples are neuronal cell bodies randomly distributed in 8 evenly separated object planes across a 350 μm DOF. (a) The lateral PSFs at -175 μm, 0 μm and 175 μm depth. The peak intensity of each PSF is scaled to [0,1] for visualization, with the corresponding scaled factor noted. (b) The raw measured images generated by each PSF technique using the same sample. (c-e) Reconstruction results for (c) all 8 depths, (d) plane -175 μm, and (e) plane -25 μm. The intersection over union (IOU) between the reconstruction results and the ground truth is noted as the average value for (c) all 8 depths, (d) plane -175 μm, and (e) plane -25 μm. The ground truth images are displayed on the right. See Supplementary Fig. S11 for an enlarged version of this figure.

Summary and discussion

In summary, we propose a simple-to-implement V-shaped PSF for wide-field microscopy. This design facilitates single-shot volumetric imaging with 3D resolution across an extended DOF. The PSF exhibits a spatially confined and depth-dependent shape throughout the extended DOF and could be generated by replacing the conventional tube lens with a pair of axicon lenses. We designed and simulated the PSF for a 10× magnification microscope and reconstructed the images at multiple depths over a DOF of 350 μm through deep neural networks. Compared with many other 3D wide-field imaging techniques through PSF engineering, our design has a large DOF. In particular, our optical setup is simple by integrating a pair of axicon lenses in the collection path and does not require complex implementation such as that in the self-bending PSF technique[19] nor complex phase mask design such as that in tetrapod PSF[17]. Our simulation showed that the deep neural network can recover high-quality 3D object information from the single image acquisition. Compared with other popular PSFs engineered for 3D imaging, the V-shaped PSF presented an excellent 3D reconstruction quality with a high IOU between the reconstructed objects and the ground truth over an extended DOF.

The single-shot 3D wide-field microscope with V-shaped PSF could be equipped with wide-field volumetric illumination or integrated with multiplane light-sheet illumination where a set of selected planes are illuminated[27]. It holds great promise for applications where it is desirable to record 3D biological structures or processes at high speed, such as *in-vivo* neuronal imaging in transparent samples (e.g., zebrafish larvae[28]), or macroscopic cleared samples[29]. Instead of imaging the sample plane by plane, multiple planes or the entire volume could be imaged together. This greatly increases the data acquisition speed, which is significant for studying fast biological dynamics or imaging macroscopic-scale samples in high-resolution.

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Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

Disclosures. The authors declare no conflicts of interest.

See Supplement 1 for supporting content.

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